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16 November 1956
SUPERSEDING
MIL-C-19113(SHIPS)
15 November 1955

MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING, ELECTRIC MOTOR DRIVEN, NAVAL SHIPBOARD LOW PRESSURE AIR SERVICE

1. SCOPE

1.1 Scope. - This specification covers stationary, base mounted, air compressors for shipboard service, driven by electric motors.

1.2 Classification. - Compressors shall be of the following classes and sizes:

1.2.1 Classes. - Compressors shall be of the following classes, as specified (see 6.1):

Class CC - Combustion control service.
Class R - Non-high impact shock construction.
Class S - High-impact shock construction.

1.2.2 Sizes. -

1.2.2.1 Class CC compressors. - Class CC compressors shall be of one of the following sizes as specified (see 6.1):

25 c.f.m. of free air delivered at 100 p.s.i.g.
50 c.f.m. of free air delivered at 100 p.s.i.g.

1.2.2.2 Class R and class S compressors. - Class R and class S compressors shall be of the following sizes, as specified (see 6.1):

Size	Maximum working pressure	Approximate free air capacity, delivered	Motor hp.
	P.s.i.	C.f.m.	
A	50	300	60
B	125	200	60
C	150	50	15
D	150	100	30
E	200	250	100
F	300	150	60
G	400	35	15

FSC 4310

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2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular (General Purpose).
QQ-A-561 - Aluminum Alloy 2S; Plate and Sheet.
QQ-C-576 - Copper Plates, Rolled Bars, Sheets, and Strips.
QQ-I-652 - Iron Castings, Gray.
QQ-S-763 - Steel, Corrosion-Resisting: Bars and Forgings (Except for Reforging).
QQ-S-766 - Steel, Corrosion-Resisting: Plates, Sheets, Strips and Structural Shapes.
GGG-P-781 - Pullers, Bushing, Bearing, Gear Wheel Hub and Cylinder Sleeve, Installing and Removing.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

JAN-P-100 - Packaging and Packing for Overseas Shipment - General Specification.
JAN-P-104 - Packaging and Packing for Overseas Shipment - Crates, Sheathed, Wood, Nailed.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Flexible (Waterproofed).
JAN-P-127 - Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
MIL-B-233 - Boxes, Repair Parts.
MIL-T-856 - Thermometer, Self-Indicating Liquid-in-Glass (Navy Type).
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-G-858 - Gray Iron Alloy; Castings (Scale-Resisting).
MIL-S-867 - Steel Castings, Corrosion Resisting Austenitic.
MIL-Z-883 - Zinc; Plates, Rods, Sheets, and Strips.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-B-892 - Bronze, Phosphor, Rolled or Drawn, Bars, Plates, Rods, Sheets, and Strips.
MIL-N-894 - Nickel-Copper Alloy Bars, Plates, and Other Wrought Forms.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application Tests for.
MIL-A-958 - Aluminum Base Alloy, Permanent Mold Castings (Marine Service).
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-B-994 - Brass, Naval, Wrought.
JAN-S-1160 - Steel, Spring, Round or Square, (Including Rectangular Sections Having Thicknesses Greater Than Half Their Widths).
MIL-T-1368 - Tubing, Nickel-Copper-Alloy, Seamless and Welded.

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MIL-C-2174 - Controllers, Motor Starters and Master Switches-Direct-Current, Naval Shipboard Use.
MIL-C-2212 - Controllers and Master Switches, Alternating-Current (Naval Shipboard Use).
MIL-C-3769 - Crates, Intermediate, Sheathed, Wood, Nailed (for Maximum Net Loads of 3000 Pounds).
MIL-T-15005 - Tubes, Copper Nickel Alloy (Condenser and Heat Exchanger).
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-15071 - Technical Manuals, for Mechanical and Electrical Equipment.
MIL-S-15083 - Steel: Castings.
MIL-P-15130 - Paint, Outside, Haze-Gray, No. 27 (Formula No. 5H) (for Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-T-15164 - Thermometer, Self Indicating, Liquid in Glass; Industrial.
MIL-C-15726 - Copper-Nickel-Alloy: Bars, Plates, Rods, Sheets and Strips.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-B-15939 - Bronze, Aluminum: Wrought.
MIL-T-15942 - Thermometers, Distant-Reading, Indicating-Dial.
MIL-B-16033 - Bronze, Aluminum: Castings.
MIL-S-16113 - Steel Plate, Hull and Ordnance, Structural, Black (Uncoated) and Zinc-Coated (Galvanized).
MIL-B-16261 - Bronze, Bearings: Castings.
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts (Includes Associated Repair Parts).
MIL-T-16420 - Tube, Copper-Nickel (70-30 and 90-10), Seamless and Welded.
MIL-B-16540 - Bronze, Phosphor: Castings.
MIL-B-16541 - Bronze, Valve: Castings.
MIL-M-16576 - Metal, Gun: Castings.
MIL-G-16983 - Gage, Pressure, Dial Indicating.
MIL-M-17060 - Motors; Alternating Current, Integral HP (Shipboard Use).
MIL-A-17129 - Aluminum Alloy Casting; Sand, Resistant to Salt-Water.
MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No-Go).
MIL-F-17292 - Format for List of Repair (Spare) Parts for Shipboard Mechanical and Electrical Equipment.
MIL-L-17331 - Lubricating Oil, Turbine (Noncorrosive).
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard.
MIL-S-17509 - Steel Castings, Austenitic, Chromium-Nickel, Low Magnetic Permeability.
MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.

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MILITARY (cont'd.)
MIL-S-17996 - Steel Sheet and Other Wrought Forms, Corrosion-Resisting Austenitic (Low Magnetic Permeability) Chromium-Manganese-Nickel.
MIL-G-18997 - Gages Pressure, Dial Indicating, Bourdon Tube, Circular Face.
MIL-S-20166 - Steel: Bars and Shapes (for Hull Construction) (Including Material for Drop and Miscellaneous Forgings).

NAVY DEPARTMENT
General Specifications for Inspection of Material.
42B10 - Bearings, Roller.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships in the U. S. Navy.

DRAWINGS

BUREAU OF SHIPS
B-214 - Root Connections for Attaching Piping.
5000-S4823-841055 - Check-Off List for Zincs in Machinery Installations Instructions for Preparation of Form.
5000-S4823-F-841337 - Unions, Silver Alloy Brazing, for Oil, Water, Gas or Oxygen.
5000-S4800-841177 - Gage Connections.

PUBLICATIONS

MILITARY
JPI-6 - Packaging and Packing of Spare Parts for Internal Combustion Engines.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION, INC.
Motor Freight Classification Rules.

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc. 1424 16th St. N.W., Washington 6, D. C.)

COMPRESSED AIR AND GAS INSTITUTE
Compressed Air Handbook, Second Edition.

(Application for copies should be addressed to the Compressed Air and Gas Institute, 122 West 42nd, New York, New York.)

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

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NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H-28 - Screw-Thread Standards for Federal Services.

(Copies of Handbook H-28 may be obtained upon application, accompanied by money order, coupon or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Qualification. - Class CC and class S compressors furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Definitions. -

3.2.1 Actual free air capacity. - Actual free air capacity as used in this specification refers to the quantity of air, expressed in cubic feet per minute at intake conditions of temperature and pressure, which is actually delivered.

3.3 General requirements. -

3.3.1 Assembly. - The equipment shall consist of compressor and driving motor mounted on a common bedplate, with such other items and accessories as may be required for the satisfactory operation of the compressor, including the following:

Filter and silencer, air intake.
Intercooler.
Aftercooler.
Gages, pressure.
Switch, pressure.
Motor controller.
Valves, drain (air, water, oil).
Valves, relief (air, water, oil).
Valves, unloading.
Thermometers.
Moisture separators.
Valve, check, air discharge.
Temperature control device, automatic.

3.3.1.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain air compressors of such design that they will operate over a long period of years with a minimum of servicing. Where wear or corrosion is unavoidable the parts subjected to such wear or corrosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum. The design and construction of all compressors shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion.
(d) Economy.
(e) Satisfactory operation when inclined as follows:
    (1) Up to 15 degrees in any direction (permanently inclined).
    (2) With the ship rolling up to 45 degrees from the vertical.
    (3) With the ship pitching 10 degrees up and down from the normal horizontal plane.
(f) Minimum space and weight (see 6.1).

3.3.1.2 The designer shall take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. For example, parts shall be so designed that it is impossible to replace them in a wrong position; adjacent studs shall be of the same length, or of large difference in lengths, so that a mistake will not easily occur; serrated surfaces shall not be used for accurate positioning, since the serrations are easily damaged; large wrench areas shall be avoided where strong wrenching would cause damage.

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3.3.1.3 Special requirements for class CC compressors. - Special requirements for class CC compressors shall be as follows:

3.3.1.3.1 The design of class CC compressors shall be such as to minimize either oil or oil fractions in the discharge from the compressors. To this end hot spots shall be avoided in cylinder heads and valves. Oil in the air discharged from the compressors shall not exceed 5 ounces (by weight) per 50 cubic feet of free air delivered per minute in 100 hours of operation (see 4.3.6.1.5).

3.3.1.3.2 The duty required for class CC compressors is 24 hours per day continuous service.

3.3.2 Standard product. - All equipment shall be essentially the product of the manufacturer wherever practicable so that prompt and continuing service and delivery of repair parts may be assured.

3.3.3 Unless otherwise specified in the contract or order, the following design conditions shall be assumed:

(a) Ambient and intake air temperature - 122° Fahrenheit (F.).
(b) Intake air pressure - 14.7 pounds per square inch (p.s.i.) absolute.
(c) Relative humidity - 50 percent.
(d) Maximum permissible temperature of air leaving the aftercooler - 100°F. for water cooled units, 162°F. for air cooled units.
(e) Temperature of cooling water at inlet - 85°F.
(f) Kind of cooling water - sea (salt).

3.3.4 Drive. - Multi-"V" belt type drive is preferred although direct connected motor drives will also be considered. If belt driven, suitable means shall be provided for tightening and adjusting the belts. Protective guards shall be provided for all exposed moving parts.

3.3.5 Rating. - The capacity and pressure rating of compressors shall be as specified (see 1.2.2). It shall be possible to satisfactorily operate each compressor at any pressure up to and including the maximum working pressure specified.

3.3.6 Shockproofness. -

3.3.6.1 All class CC and S compressors shall be designed to withstand shock due to firing of the ship's own armament and noncontact underwater explosions of near-miss aerial bombs, torpedoes, and mines. Every attempt shall be made to design the equipment to resist shock as specified by graphs A, B, and C, shown on figure 1, it being understood that these shocks are applied at the footings of the component. Resilient mountings shall not be used unless bureau or agency concerned approval of the mount and its proposed application is given.

3.3.6.2 Holding-down bolts, supporting feet or legs, and subbases shall be designed to resist shock by using the accelerations specified on the graph shown on figure 1 for determining shock forces. The determination of stresses resulting from these forces is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. Where two or more components are installed on a common rigid subbase, the feet and holding-down bolts, as well as the subbase, shall be designed for an acceleration based on the total weight of the assembly.

3.3.6.3 The design of all complete class CC and S compressor units shall be such that they are capable of passing the high-impact shock test specified in Specification MIL-S-901.

3.3.6.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

3.3.7 Compressor. -

3.3.7.1 All compressors shall be two stage design, except that size A units may be single stage, and size G may be three stage.

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3.3.7.2 Piston speeds of class R and S compressors shall not exceed 900 feet per minute. The rotating speed of the crankshaft of class QC compressors shall not exceed 500 revolutions per minute (r.p.m.).

3.3.7.3 Air passages shall be unrestricted.

3.3.7.4 The compressor gage board, drain valves and all features of an operational nature shall be located on or shall be operable from one side of the compressor.

3.3.7.5 All parts subject to wear or service failure shall be replaceable. Cover plates for access openings shall be oil and dust tight.

3.3.7.6 Noise. - The compressor shall be designed to operate as quietly as possible.

3.3.7.7 The crank case frame, housing, cylinders, and other fixed parts or attached components of the assembly shall be rigidly secured and held in alignment by dowel pins, fitted bolts, keys, rabbeted or tongue and groove joints, or other suitable means.

3.3.7.8 Air cylinders and heads. - So far as practicable hand holes shall be provided for access to water jacket spaces. Cylinder liners if provided shall be held rigidly in place. Unless otherwise approved by the bureau or agency concerned, the liners shall be counterbored slightly at the top ends to permit the piston rings to overide the working bore to avoid the formation of shoulders. The design and machining of cylinders and liners shall be such that the complete piston assembly can be removed, and replaced. To this end all shoulders where piston rings may catch shall be avoided and liners shall be properly beveled.

3.3.7.9 Bedplates. - The bedplates shall be continuous under the compressor and its driving unit. They shall be machined to receive the intermediate frames or housing, or may be integral therewith. All top and bottom bearing surfaces shall be machine finished. Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship, without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the supporting structures onboard ships cannot cause misalignment of compressors.

3.3.7.10 Housing. - The housing shall consist of cast frames, cast frames and forged steel tie rods, or structural steel welded assembly which shall support the cylinders. The crankshaft and the housing shall be wholly enclosed to prevent the splashing of oil. Access openings with covers shall be provided for observation and adjustment of the bearings, and where necessary breather openings fitted with screens shall be provided.

3.3.7.11 Air filter. - Provisions shall be made for cleaning the inlet air and silencing the inlet air noise. These functions may be performed by a combination cleaner silencer, or separate devices for each function. Provision shall be made for replacement of cleaner elements. Inlet connection shall be flanged for possible attachment of an external pipe to conduct clean air to the compressor from remote location. These devices shall be as compact as practicable and shall be mounted on the compressor.

3.3.7.12 Lubrication. -

3.3.7.12.1 An adequate and self-contained lubricating system shall be provided for the main bearings, crankpin bearings, wrist-pin bearings, gears and pinions. A suitable oil sump or reservoir shall be provided. If a force feed lubricating system is provided the oil shall be drawn from the reservoir by one or more pumps which are positively driven from the compressor crankshaft or motor shaft. The arrangement of reservoir and pumps shall be such that oil will not be lost from the reservoir nor the pump lose suction due to roll or pitch of the ship. The lubricating oil used shall be in accordance with Specification MIL-L-17331.

3.3.7.12.2 The oil reservior shall be provided with suitable fittings for filling and draining. The drain fitting shall be arranged so that the reservoir can be completely drained when the compressor base is level. An oil level gage, marked to show the high, low, and normal oil levels, shall be fitted. Gage glasses, if used, shall be renewable without draining the oil from the reservior.

3.3.7.12.3 If a forced feed system is provided the oil shall be properly strained before being delivered to the parts to be lubricated. The strainer shall be fitted on the discharge side of the lubricating oil pump. Crankcase oil temperatures shall not exceed 180°F. Oil coolers (if provided) shall be designed to withstand pressure on the cooling water side of 150 p.s.i., unless otherwise specified in the contract or order. Suitable means shall be provided for regulating the pressure of the oil.

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3.3.7.13 Piping and valves. - All water, air and oil connections to the compressor shall be readily accessible. Pockets where oil or water may collect shall be avoided, or if unavoidable, shall be arranged for convenient drainage. All compressor water connections to external shipboard piping systems shall be flanged. Flange and bolting standards shall be in accordance with Drawing B-214. Air foot connections shall be in accordance with Drawing 5000-S4623-F-641337. All air, water and oil piping on the unit shall be made up with connections so as to be readily removable for overhaul of the unit. Threaded pipe connections shall be reduced to an absolute minimum on internal piping. Compression type or flanged fittings shall be used wherever possible. Piping shall be so fitted as to avoid necessity of removing any part of same for access to and removal of crankcase and valve covers. All relief valves shall have the discharge directed in such manner as not to discharge in the face of operating personnel.

3.3.7.14 Threaded parts. - All threaded parts shall be in accordance with Handbook H-28. Use of studs shall be avoided.

3.3.7.15 Jacking device. - Suitable provision shall be made for jacking the compressor over, if it is impractical to turn the machine by hand.

3.3.7.16 Cooling system. -

3.3.7.16.1 General. - Each compressor shall be provided with an efficient cooling system including an intercooler and an aftercooler. All class CC compressors shall be water cooled with ample water jackets for cylinders, heads and coolers. The method of cooling of class R and S compressors shall be as specified (see 6.1). The temperature of air discharged from any stage of any compressor shall not exceed 400° F. The final air temperature shall not exceed the values indicated in 3.3.3.

3.3.7.16.2 Air cooled compressors. -

3.3.7.16.2.1 Unless otherwise specified in the contract or order, all air cooled compressors shall be completely air cooled, requiring no connections from the ships' water system. Cooling air shall be supplied by a fan integral with the compressor.

3.3.7.16.2.2 When so specified (see 6.1 and 6.3) air cooled compressors may have air cooled cylinders and heads with water cooled intercoolers and aftercoolers.

3.3.7.16.2.3 Cylinders and heads of air cooled compressors shall have ample radiating surface in the form of fins cast integral with these parts. Air cooled inter and aftercoolers shall be of seamless copper or copper alloy with copper or copper alloy radiating fins. Water cooled inter and aftercoolers shall conform to 3.3.7.16.3.5.

3.3.7.16.3 Water cooled compressors. -

3.3.7.16.3.1 All parts of the cooling system of water cooled compressors shall be of corrosion resisting material and shall be designed for the use of salt water supplied from the ship's flushing or firemains, or, when specified in the contract or order, from an attached or independent circulating water pump. The cooling system shall be designed for working pressure of 50 p.s.i. on the cooling water inlet and shall be capable of withstanding a maximum pressure of 150 p.s.i. Provision shall be made for completely draining all parts of the circulating water system.

3.3.7.16.3.2 Consideration will be given to a closed cooling system employing liquids other than water, providing it can be demonstrated that there will be a saving thereby in cost or weight, or other definite advantages.

3.3.7.16.3.3 The cooling water system shall be so arranged that the quantity of cooling water supplied to the cooler (if provided) may be regulated without affecting the quantity of cooling water supplied to the remainder of the system.

3.3.7.16.3.4 Provision shall be made for positive and automatic control of the cooling water so that the water is turned on when the compressor is running and turned off when the compressor is not running (see 3.5.2.2).

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3.3.7.16.3.5 Coolers. - Inter and aftercoolers shall be of the straight-tube type. Consideration will be given to the use of coil-type coolers. Above pressures of 250 p.s.i., the air shall flow through the cooler tubes, with water flowing over and around the tubes. Cooler tubes shall be rolled or welded into tube sheets or headers at both ends. Provision shall be made for the expansion of tube nests. Stay rods, if and as necessary shall be provided for support of tube sheets and air heads. Cooling coils, if used, shall be rigidly tied together by strips brazed to each coil turn. The complete coil assembly shall be supported in the cooler in a manner that will prevent vibration of the coil. Each cooler tube nest shall be a self-contained unit so that it may be easily removed from the compressor as a unit for inspection and cleaning. Suitable water and air relief valves shall be fitted on each cooler, except that a water relief valve on the intercooler will not be required. Unless otherwise specified in the contract or order, water relief valves shall be set at 155 p.s.i. Salt water cooling systems shall be protected by zincs.

3.3.7.16.3.6 Lubricating oil coolers and jacket water coolers furnished as separate accessories shall conform to Specification MIL-C-15730. Such coolers fabricated as an integral part of the compressor shall conform to the basic design requirements of Specification MIL-C-15730.

3.3.7.16.3.7 Zinc. - The upstream side of the inter and aftercoolers shall be fitted with zincs of a design similar to any one of those shown on Drawing 5000-S4823-841055. The zincs shall have as large a surface area as is practicable for maximum electrolytic protection without causing excessive turbulence The zincs shall be easily removable for inspection and replacement. Zincs shall be special high grade material as specified in Specification MIL-Z-883.

3.3.7.17 Unloading. - All compressors shall be fitted with automatic unloading valves so that the compressor will always be started up in the completely unloaded condition, that is, removal of all load except inertia and friction load of the machine (see 3.5.2.2).

3.3.7.18 Gages and gage board. - All gages shall be 3-1/2 inch size of the dial indicating type as shown in Specification MIL-G-18997. All gages and instruments shall be mounted on a common gage board. The gage board shall be provided with brackets for mounting on the compressor. Mounting brackets shall be of a type which can be removed from the compressor when the gage board is bulkhead mounted. Pressure gages shall be provided to indicate the air-delivery pressure from each stage of compression, the pressure in the circulating water system and the lubricating oil pressure. All gages on the gage board shall be suitably identified on an identification plate located on the gage board adjacent to each gage. The air gages for each stage shall be marked to show the normal designed working pressure, and such working pressures shall be indicated on drawings and in technical manuals.

3.3.7.19 Connecting rods. - The crank end of connecting rods shall be arranged for renewal of crankpin bearings. Bolts and nuts for securing crankpin bearings shall be positively locked. The end of connecting rods connected to crossheads or pistons shall be fitted with renewable bearings. Provision for adjustment shall be made as required for proper operation and maintenance of the compressor. Any take-up bolts and nuts shall be positively locked. Upper end of connecting rods shall be of one-piece construction.

3.3.7.20 Crankshaft. - The crankshaft shall be a single forging or casting. The cranks shall be arranged to give the best possible running balance. Counterweights shall be fitted if necessary. The design of the shaft shall be such as to permit convenient use of a portable tachometer.

3.3.7.21 Flywheels. - Compressors shall be equipped with a flywheel of sufficient size to provide smooth running of the compressor and motor. Flywheels shall be keyed to the crankshaft and shall be pressed or securely clamped thereon. A separate flywheel will not be required if ample flywheel effect is otherwise provided.

3.3.7.22 Bearings. -

3.3.7.22.1 Main bearings shall be readily renewable. The main bearing shall preferably be of the tapered roller or spherical type in accordance with Specification 42B10 or FF-B-185 which shall be so arranged as to take up the thrust due to roll and pitch of the vessel. Consideration will be given to the use of deep groove or angular contact type ball bearings in accordance with Specification FF-B-171. Access openings with covers shall be provided for observation and adjustment of bearings, and, where necessary, breather openings with screens shall also be provided.

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3.3.7.22.2 Crankpin bearings shall be thin wall or thin babbitt precision type insert bearings.

3.3.7.22.3 Wrist-pin bushings and crosshead bearings shall be bearing bronze, grade I or II, in accordance with Specification MIL-B-16261.

3.3.7.23 Moisture separators - Efficient separators shall be provided for removing free moisture and oil from the air upon delivery from each intercooler and aftercooler. All separators shall be fitted for blowing out accumulations of water and oil. Drain provisions shall be accessible for operation from the operating or gage board side of the unit, insofar as practicable.

3.3.7.24 Thermometers. - Thermometers shall be provided for the measurement of air temperatures before and after each stage of compression and in the discharge beyond the aftercooler. Thermometers shall also be provided for the measurement of oil temperature and for the measurement of cooling water outlet temperatures in water cooled systems. Dial type thermometers, Specification MIL-T-15942, are preferred, other type thermometers as specified in Specifications MIL-T-855 and MIL-T-15164 will however also be considered. Plastic lenses shall be supplied in lieu of glass where practicable. Thermometer ranges shall be limited to the following: 20° to 180°F., 30° to 240°F., 70° to 400°F or 100° to 550°F.

3.3.8 Welding. - Welding and allied processes shall be in accordance with Standard MIL-STD-278.

3.3.9 Painting. - All surfaces to be painted shall be thoroughly cleaned and free from oil, grease, mill scale, rust and loose weld beads. All external nonworking ferrous and aluminum surfaces shall be painted with one coat of zinc chromate primer in accordance with Specification JAN-P-735, and two coats of haze-gray paint in accordance with Specification MIL-P-15130.

3.3.10 Identification plates. -

3.3.10.1 Identification plates shall be furnished on each compressor and shall be type A or B in accordance with Specification MIL-I-15024, except that identification plates of plastic or aluminum are not acceptable.

3.3.10.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.3.10.3 Identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
    (1) Capacity.
    (2) Pressure in p.s.i.
    (3) Speed of shaft in r.p.m.
    (4) Brake horsepower.
    (5) Working pressures of each stage.
    (6) Lubricating oil pressure.
    (7) Cooling water pressure required.
    (8) Set pressures of each relief valve.
    (9) Any special data vital to the unit.
(f) Contract number (and item number of multiple unit orders).
(g) Bureau of Ships stock number (space for number should be provided even though number may not be assigned).
(h) Section for inspector's stamp.

3.3.10.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification listed in section 2.

3.4 Materials. -

3.4.1 General. - All material used in the construction of compressors and associated equipment shall be of the best quality for the purpose intended.

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3.4.2 The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the bureau or agency concerned.

3.4.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the compressor, The use or substitution of manufacturer's material shall be subject to approval by the bureau or agency concerned and shall be clearly indicated in a list submitted with the master drawings.

3.4.4 Class CC and S compressors. - Material used in the construction of class CC and S compressors and associated equipment shall be strictly in accordance with the material specifications listed in table I.

Table I - Material.

Application	Material	Specification
Bedplates	Cast steel	MIL-S-15083 or MIL-S-17509
	Structural steel, welded	MIL-S-20166; MIL-S-16113
	Corrosion resisting steel	QQ-S-763; QQ-S-766 or MIL-S-17996
	Aluminum bronze	MIL-B-15939
Frames and housings	Cast steel	MIL-S-15083 or MIL-S-17509
	Structural steel, welded	MIL-S-20166; MIL-S-16113
	Nodular graphitic iron	MIL-I-17166
	Gun metal	MIL-M-16576
	Valve bronze	MIL-B-16541
For class R only	Cast aluminum	MIL-A-17129, class 4 - HT2
Cylinders, air; Cylinder heads For class R only	See note 2 Gun metal Valve bronze Cast aluminum	MIL-M-16576 MIL-B-16541 MIL-A-17129, class 4 - HT2
Cylinder liners	Hard close grained cast iron	MIL-G-858, QQ-I-652
Crankshaft; Connecting rods, caps, and bolts	Forged steel, class An or Bs	MIL-S-890
	Corrosion testing steel	QQ-S-763 or MIL-S-17996
	Aluminum bronze	MIL-B-15939
	Nodular iron	MIL-I-17166
Wrist pins	Forged steel, class Ac or An, hardened and ground	MIL-S-890
	Corrosion resisting steel	QQ-S-763 or MIL-S-17996

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Table I - Material (cont'd.).

Application	Material	Specification
Wrist pin bushings	Bearing bronze, grade I or II	MIL-B-16261
Pistons	High test gray iron Cast iron Aluminum Aluminum bronze	QQ-I-652 MIL-A-958 MIL-B-16033
Flywheel	Cast steel, class B Nodular graphite iron Fabricated steel, welded	MIL-S-15083 MIL-I-17166 MIL-S-20166, MIL-S-16113
Piston rings	Hard close grained cast iron preferably centrifugally cast	
Air valves	Corrosion-resisting steel, class FS-304 or FS-410	QQ-S-763 or MIL-S-17996
Air valve seats	Corrosion-resisting steel, class FS-304 or FS-302 Corrosion-resisting steel, class 1 or 2, respectively Bronze Bronze	QQ-S-763 or MIL-S-17996 MIL-S-867 or MIL-S-17509 MIL-B-16540 MIL-B-892
Air valve springs	Spring steel, cadmium plated Corrosion-resisting steel	JAN-S-1160 QQ-S-763 or MIL-S-17996
Intercooler and after-cooler Tubes, seamless	See note 1 Copper nickel alloy Nickel copper alloy	MIL-T-15005 MIL-T-1368
Intercooler and after-cooler tube sheets or headers	See note 3 Naval brass Nickel copper alloy	MIL-B-994 MIL-N-894
Intercooler and after-cooler Air heads and moisture separators	See note 3 Aluminum bronze, class 1 Gun metal Valve bronze	MIL-B-16033 MIL-M-16576 MIL-B-16541
Intercooler and after-cooler Waterheads	Gun metal Valve bronze	MIL-M-16576 MIL-B-16541

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Table I - Material (cont'd.).

Application	Material	Specification
Intercooler and after-cooler shells	See note 3 Seamless drawn tubing same as tubes, or Gun metal Valve bronze	 MIL-T-16420 MIL-M-16576 MIL-B-16541
Intercooler and after-cooler shell baffles	Naval brass, or same material as shell CU-NU	MIL-B-994 MIL-C-15726
Intercooler and after-cooler	Naval brass	MIL-B-994
Stay rods	Nickel copper alloy	MIL-N-894
Intercooler and after-cooler gaskets (air side)	Copper, soft iron, or dead soft aluminum sheet	QQ-C-576 QQ-A-561
Sheaves	Cast steel Cast iron Valve bronze Nodular graphitic iron Fabricated steel, welded	MIL-S-15083 or MIL-S-17509 MIL-G-858 MIL-B-16541 MIL-I-17166 MIL-S-20166, MIL-S-16113

NOTES:

1. Seamless tubes for intercoolers and aftercoolers shall be 90-10 copper-nickel.
2. Air cooled cylinders and heads, and noncorrosive liquid cooled cylinders and heads may be of nodular graphitic iron in accordance with Specification MIL-I-17166.
3. Intercooler and aftercooler shells, air heads, and separators which are not in contact with water may be of cast steel, Specification MIL-S-15083 or MIL-S-17509, or of nodular graphitic iron, Specification MIL-I-17166.

3.4.5 Class R compressors. - Material used in the construction of class R compressors and associated equipment shall be identical to or the commercial equivalent to materials indicated in table I. In general, cast iron will not be permitted except for such items as piston rings or other nonstructural parts where experience indicates the use of cast iron is satisfactory. Cast steel or nodular cast iron shall be used in lieu of cast iron.

3.4.6 When the contract or order specifies that the compressor units shall be nonmagnetic the materials of construction shall be selected such that the magnetic permeability of the compressor component unit (less motor and controller) after fabrication will be 2.0 or less measured by a permeability indicator conforming to Specification MIL-I-17214.

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3.4.3.1 In order to achieve the desired low permeability after fabrication it is advisable to choose materials which have a permeability prior to fabrication of 1.2 or less when materials whose permeability is affected by fabrication (CRES) are used.

3.5 Electrical equipment. -

3.5.1 Motors. - All motors shall be in accordance with Specification MIL-M-17413 for direct current (d.c.) motors, or Specification MIL-M-17060 for alternating current (a.c.) motors, and the following:

(a)	Service	Navy A for class CC and S compressors; Navy C for class R compressors.
(b)	Voltage and current characteristics	As specified (see 6.1).
(c)	Enclosure	Dripproof protected.
(d)	Ambient temperature	50°C.
(e)	Duty	Continuous.
(f)	Speed	Constant.
(g)	Winding	A.c. squirrel cage induction. D.c. stabilized shunt.
(h)	Insulation	Class A or C.
(i)	Bearings	Ball.

3.5.1.1 The horsepower rating of each motor shall be in accordance with Navy standard motor ratings, but not less than 1.05 times the brake horsepower of the driven compressor under rated conditions of capacity and pressure.

3.5.1.2 Radio noise. - The requirements in the motor specifications specified in section 2 for minimizing the generation of radio interference shall apply to both service A and service C motors.

3.5.2 Controllers and master switches. - All controllers and master switches shall be in accordance with Specification MIL-C-2212 for a.c. equipment, or Specification MIL-C-2174 for d.c. equipment, and the following:

Enclosure	Dripproof.
Operation	Magnetic.
Ambient temperature	50° Centigrade (C.).
Insulation	Class B (class A for coils only).
Performance	See 3.5.2.1.
Control function	Motor starting.
Type	A.c. across line. D.c. resistor, inductive time lag acceleration.
Protection	Overload, low voltage (l.v.p.).
Duty	Same as motor (see 3.5.1).
Master switches required	See 3.5.2.1.
Master switch enclosures	Spraytight (remote) dripproof (local).
Master switch location	Remote or local.

3.5.2.1 Each controller shall be provided with local "ON, OFF, TEST" push buttons, a two-position selector switch marked "START-STOP" and "CONSTANT SPEED", and a pressure switch. The push buttons shall be of the momentary contact type.

3.5.2.1.1 With the selector switch in the START-STOP position, pressing the "ON" push button shall place the compressor in operation under automatic control of the pressure switch. When the air pressure in the discharge line is below the value set on the pressure switch, the pressure switch shall close and start the compressor. The compressor shall continue to run until the pressure rises to the value required to open the pressure switch causing the controller to stop the compressor. This start-stop operation shall continue in effect until the "OFF" push button is pressed.

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3.5.2.1.2 With the selector switch in the "CONSTANT SPEED" position the compressor shall start when the "ON" push button is pressed (regardless of air pressure in the discharge line) and shall continue to run until the "OFF" push button is pressed. With this connection the compressor shall automatically unload when the pressure builds up to a set value; and when the air pressure has fallen to a lower set value the unloader shall reclose permitting the compressor to build up pressure again. Unloading may be accomplished by mechanical means, or by a magnetic unloader controlled by the pressure switch.

3.5.2.1.3 The "TEST" push button shall be arranged to "jog" or "inch" the motor. Overload protection shall not be by-passed by either the "TEST" or "ON" push buttons. Emergency run feature shall not be provided.

3.5.2.1.4 The pressure switch shall be service A, in accordance with Specification MIL-C-2174 or MIL-C-2212, as applicable. The pressure switch shall be of an adjustable type, suitable for the maximum pressure specified for the compressor and with a minimum differential of not more than 6.7 percent of the compressor rating.

3.5.2.2 Cooling system valves (see 3.3.7.16.3.4) if electrically operated, and unloading valves (see 3.3.7.17) if electrically operated shall conform to all the applicable requirements for controllers as specified in 3.5.2, except that enclosures shall be watertight (or better). Valves shall be tight-seated and all internal parts in contact with water shall be nonferrous material except where ferrous parts are necessary for the magnetic circuit, and such parts shall be suitably nickel-plated for protection against corrosion. Cooling system valves shall be provided with a means of manually opening the valves in case of coil failures (see 3.3.7.13).

3.6 Repair parts and tools. -

3.6.1 Repair parts and tools shall be furnished and processed in accordance with Specification MIL-R-15137.

3.6.2 Onboard repair parts. -

3.6.2.1 Each set of repair parts required by the contract or order shall contain the parts listed in table II, plus any additional parts essential in successful servicing of the compressors.

Table II - Onboard repair parts.

Parts as applicable to compressors furnished	Quantity of parts furnished per set of repair parts					
	Number of compressors having identical parts					
	1	2	3	4	5	6 or more
Bearings, ball or roller (except main) sets	1	2	3	4	5	6
Bearings, crank pin sets	1	1	1	2	2	2
Bearings, main sets	1	1	1	2	2	2
Bearings, wrist pin sets	1	1	1	2	2	2
Belts, complete sets	2	3	4	5	6	7
Bolts, nuts, washers, connecting rod, (crank end), complete sets	1	1	1	2	2	2
Bushings, each size and type	1	1	1	2	2	2
Couplings, flexible, complete, sets	1	1	1	2	2	2
Gages, oil level	1	1	1	2	2	2
Gages, pressure, sets	1	1	1	2	2	2
Gaskets, sets	1	2	3	4	5	6
Packing, sets	1	1	1	1	1	1

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Table II - Onboard repair parts (cont'd.).

Parts as applicable to compressors furnished	Quantity of parts furnished per set of repair parts						
	Number of compressors having identical parts						
	1	2	3	4	5	6 or more	
Pins, wrist or crosshead, sets	1	1	1	2	2	2	
Pistons, each size	1	1	0	0	0	0	
Pumps, oil, complete	1	1	1	2	2	-	
Rings, piston, sets, each stage	1	2	3	4	5	6	
Rods, connecting, each size	1	1	0	0	0	0	
Springs, sets	1	2	3	4	5	6	
Strainers, oil	1	1	1	2	2	-	
Tube coils, intercooler and aftercooler, or removable tube nets, sets	1	1	1	2	2	-	
Tubes, intercooler and aftercooler, where easily renewable percent of each size and type	10	10	10	20	20	20	
Valves, discharge for all stages, less removable seats, springs, guards, sets	1	2	3	4	5	6	
Valves, discharge, for all stages, complete with seats, guards, springs, etc., sets	1	1	1	2	2	2	
Valves, inlet for all stages, less removable seats, springs, guards, sets	1	2	3	4	5	6	
Valves, inlet for all stages, complete with seats, guards, springs, etc., sets	1	1	1	2	2	2	
Working parts for automatic control devices, sets	1	1	1	2	2	2	
Bearings, crosshead, sets (or crossheads, sets if bearings integral)	1	1	1	2	2	-	

3.6.2.2 The quantity of each particular item in a set of repair parts is based on the number of compressors having that item which are installed onboard a vessel even though the compressors may not be identical. For example, a set of repair parts may be intended to service two 50 cfm. 100 p.s.i. compressors and three 100 cfm. 100 p.s.i. compressors; if an identical gasket is used on both sizes of compressor the quantity of gaskets in a repair parts set shall be determined from the "5 compressor" column of table II. The quantity of compressors installed onboard a vessel shall be specified by the bureau or agency concerned (see 6.1).

3.6.2.3 Repair parts for motors, controllers, pressure switches and other devices shall be furnished as required by the applicable specifications listed in section 2.

3.6.3 Tools and wrenches (see 6.1). -

3.6.3.1 Each set of special maintenance tools required by a contract or order shall consist of such tools as trams for setting piston clearances, gear wheel, flywheel and coupling pulling tools, and such other special tools as may be necessary for the proper repair and maintenance of the compressors. Each set of special tools shall be furnished in separate steel boxes.

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3.6.3.1.1 Note to ordering activities. - Normally the number of sets of special maintenance tools should be the same as the number of sets of onboard repair parts. However the number of sets of special tools may be increased as required by the size and organization of the ships.

3.6.3.2 Each set of special operating wrenches required by a contract or order shall consist of wrenches and tools, such as valve wrenching plates, as normally required by the compressor operators. Each set of special operating wrenches shall be furnished in separate steel boxes.

3.6.3.2.1 Note to ordering activities. - Normally a set of special operating wrenches shall be ordered for each compartment in which an air compressor is installed.

3.6.3.3 Tools which are available in the Catalog of Naval Material, General Stores Section, (copies of this catalog may be consulted in the office of the Government inspector), such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of pullers sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.6.4 Stock. - The selection, stock numbering and quantities of repair parts for stock shall be in accordance with Specification MIL-R-15137.

3.6.4.1 The cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania, unless otherwise specified in the contract or order.

3.7 Drawings. -

3.7.1 Drawings shall be in accordance with type I of Specification MIL-D-963.

3.7.2 Outline drawings shall be furnished covering complete units, including compressor, driver, bedplate and attached accessories. In addition to the data specified in Specification MIL-D-963, the center of gravity for the compressor assembly, wet, shall be shown on the outline drawing. Separate outline drawings shall be furnished for each contract or order and for each different compressor unit on a contract or order.

3.7.2.1 If the compressors and motors are close-coupled or built together so that they are not in fact separate components, then the outline drawing may be considered as a master drawing, entitled "General Arrangement and List of Reference Drawings", and all certification data applying to an individual contract or order (including electrical controller reference drawings) should be omitted from the general arrangement drawing. In this case separate certification data sheets including all data applicable to the individual contract or order shall be furnished, and the list of repair parts may be included in the certification data sheets or on a separate master drawing to be referenced by the certification data sheets. Certification data shall be on size A sheets.

3.7.3 Compressor master drawings shall include sectional assembly, and elevation drawings, and details of all parts. The sectional assembly and elevation drawings shall identify all parts and shall include a complete list of material with references to the detail drawings. The details shall be on one or more additional sheets. Size F drawings (28 by 40 inches) are preferred for sectional assembly and detail drawings.

3.7.3.1 If more than one sectional assembly or elevation drawing is required to identify all parts, or if the sectional assembly with list of material results in a drawing larger than size H (28 by 60 inches) then the list of material may be on a separate master drawing.

3.7.3.2 It is recommended that details of parts common to all or several classes of compressors be grouped together on the same sheet(s); and likewise all details applying to a single class should be grouped together on the same sheet(s).

3.7.3.3 The sectional assembly drawing shall show all assembly clearances, and shall reference the qualification test, high impact shock test, and inclination test reports, when such reports are required, and the bureau of agency letters approving those reports, except that when a general arrangement drawing (see 3.7.2.1) is used, this data may be referenced on the general arrangement drawing.

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3.7.1 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.7.5 Drawings for drivers and associated equipment shall be in accordance with the specifications for these equipments.

3.7.6 Drawings submitted with bids are not required to be in master drawing format.

3.8 Technical manuals. - Technical manuals shall be in accordance with type C of Specification MIL-T-15071, unless otherwise specified in the contract or order. A separate manual shall be furnished for each different compressor unit except in special cases when manuals covering more than one compressor may be approved. Preliminary manuals submitted for approval shall include all proposed sections complete.

3.8.1 Illustrations. - Two photographs, taken 180 degrees apart on a horizontal plane shall be included. Reduced size reproductions of approved outline, assembly and sectional drawings of the complete unit shall be included. Reduced size reproductions of approved electrical drawings and certification data shall also be included. Approved drawing excerpts and exploded views of parts or assemblies requiring detailed instruction, shall be included.

3.8.2 Distribution - Two copies of technical manuals shall be packed with each compressor unit. Additional copies of manuals shall be furnished as specified (see 6.1). If final manuals are not available at the time compressors are shipped then two copies of preliminary manuals shall be packed with each compressor unit; and the final manuals shall be delivered at a later time in accordance with Specification MIL-T-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Sampling. - Completely assembled compressors, including driving units, temperature controls, unloading valves and all equipment or fittings shall be selected by the Government inspector in accordance with table III for the tests specified in 4.3.2.

Table III - Sampling.

Number of identical units on order	Random selection
Up to 20	1 from every 5 units
21 to 100	1 from every 10 units, but not less than 4
Above 100	1 from every 20 units, but not less than 10

4.3 Tests. -

4.3.1 All tests specified herein shall be at the expense of the exhibitor unless otherwise specified or approved by the bureau or agency concerned.

4.3.2 Lot acceptance tests. - Compressors selected in accordance with 4.2 shall be tested at the place of manufacture in accordance with 4.3.4, 4.3.5, 4.3.6 and 4.3.9.

4.3.3 Qualification inspection (for class CC and class S). - Qualification tests shall be conducted at the place of manufacture for the test specified in 4.3.4 and at a laboratory designated by the Bureau of Ships for the tests specified in 4.3.5 to 4.3.9, inclusive. The capacity, maximum working pressure, and other characteristics of units for qualification tests shall be as arranged between the Bureau of Ships and the exhibitor

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4.3.3.1 Qualification tests shall be conducted to determine that performance and design of the class CC and class S compressors are in compliance with the requirements of this specification. Minor deviations in design and materials may be allowed at the discretion of the Bureau of Ships.

4.3.4 Hydrostatic test. - The various parts of each compressor shall be hydrostatically tested at the place of manufacture in accordance with the following:

(a) Cylinders, 1-1/2 times the maximum designed working pressure.
(b) Water jackets, 225 p.s.i.g.
(c) Intercoolers and aftercoolers, 225 p.s.i.g. on the water side and 1-1/2 times the maximum design working pressure on the air side.
(d) Piping and other parts not mentioned herein and subjected to pressure shall be tested to 1-1/2 times the maximum designed working pressure.

4.3.5 Electrical. - Motors and controllers shall be tested as required by the applicable specifications listed in section 2.

4.3.6 Performance tests. -

4.3.6.1 Compressor units selected in accordance with 4.2 shall be submitted to a continuous running test of 8 hours' duration at normal capacity and pressure. Test is to be run under normal electrical conditions as specified (see 6.1). Power consumption tests shall be conducted as a part of the tests. On these tests the following data shall be recorded at the start and finish of the run and at regular intervals, not exceeding 1/2 hour throughout the run.

4.3.6.1.1 Capacity test. - The measurement of delivered air shall be in accordance with the A.S.M.E. Long-Radius Low-Ratio nozzle method specified in the Compressed Air Handbook.

4.3.6.1.2 Recorded data. - Recorded data shall be as follows:

MECHANICAL

(a) Dry bulb ambient temperature.
(b) Wet bulb ambient temperature.
(c) Intake temperature (in air stream close to intake).
(d) Barometer readings.
(e) Air discharge pressure from each stage of compression.
(f) Final air pressure at discharge from aftercooler.
(g) Air temperatures before and after each stage of compression.
(h) Circulating water inlet and outlet temperatures.
(i) Quantity of cooling water in gallons per minute.
(j) Cooling water pressure drop at rated flow.
(k) Revolutions per minute of driving unit and of compressor.
(l) Weights of oil filters (see 4.3.6.1.5).

ELECTRICAL

(a) Voltage amperage and frequency.
(b) Power factor.
(c) Power input to motor.
(d) Complete data as to performance of motor including all data required to determine suitability of the electrical equipment for the specific application. This data shall be recorded primarily to demonstrate that the motor is of adequate power for the service intended.

OPERATIONAL

(a) Operation of the automatic temperature control (if provided).
(b) Operation of the automatic unloading devices.
(c) Operation of moisture separators. (Include record of quantity of oil and water drained from each separator.

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4.3.6.1.3 Computed data. - The following data shall be computed from the recorded data and shall be included on each test data record:

(a) Capacity in cubic feet of free air per minute actually delivered, computed as indicated in the Compressed Air Handbook.
(b) Brake horsepower at the compressor shaft (computed from previous test of motor).
(c) Volumetric efficiency.
(d) Overall efficiency of compressor that is, compression efficiency X mechanical efficiency. Theoretical air horsepower upon which efficiency is based may be calculated from either the adiabatic or isothermal base, but test data should indicate which is used.
(e) Amount of oil in discharged air of class CC compressors.

4.3.6.1.4 During each operational test the functioning of the compressor units shall be smooth and free from excessive heating, noise, and vibration and the performance in all respects shall meet the requirements of this specification.

4.3.6.1.5 In addition to and following the performance tests, one class CC compressor on each contract or order shall be given a continuous operating test of 100 hours duration. During this test the compressor shall discharge through an absorbent type filter of a type which will filter out all oil from the air. The filter shall be carefully weighed in a warm dry condition prior to the start of this test and again after the 100 hours run (see 3.3.1.5). The filter weights shall be included in the recorded data. During this test the compressor shall be operated under "Constant Speed" control (see 3.5.2.1.2) with the compressor unloaded not less than 5 minutes nor more than 10 minutes out of each hour. During this test the temperature of the cooling water shall not be cooler than 65° F. (see 3.3.1.3).

4.3.6.1.6 Upon completion of the performance tests, each compressor shall be disassembled sufficiently to permit inspection of all parts subject to wear or damage by running, these parts to include bearings, wrist pins, pistons and piston rings, cylinders, and air valves. To be acceptable no parts of the compressors shall show any signs of failure or excessive wear. The condition of the compressors at the conclusion of all tests shall be noted on the official test sheets.

4.3.6.2 Each compressor unit not selected in accordance with 4.2 shall be subjected to tests exactly as required by 4.3.6, except:

(a) The duration of the test need not exceed 4 hours, and data is required to be taken only at the start and finish of the run and once in between.
(b) The units need not be disassembled upon completion of the performance tests, unless excessive heating, noise, vibration, or erratic performance is noted during the tests.

4.3.6.2.1 If excessive heating, noise, vibration, or erratic performance is noted and is corrected, the 4-hour test shall be repeated. If the first correction does not completely eliminate the trouble, then after further correction the compressor shall be subjected to a complete 8-hour test before acceptance.

4.3.6.2.2 All recorded data and computation required for the 8-hour test are also required for the 4-hour test.

4.3.7 Inclination test. -

4.3.7.1 One compressor of each design and class including the driving motors, shall be subjected to the inclination tests to determine satisfactory compliance with 3.3.1.1, and the tests need not be repeated unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Neither changes in materials only of compressor construction, nor changes in the driving motor shall be considered as reasons for retesting.

4.3.7.2 The compressor unit shall be operated at rated load for not less than 30 minutes with the base tilted 15 degrees in each of the four cardinal positions, right, left, front, and rear. During the progress of these tests there shall be no excessive heating of any parts, and noise and vibration shall not exceed that of normal horizontal operation.

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4.3.7.3 The suitability of the compressor unit for operation when a ship is rolling or pitching shall be determined by inspection of the design.

4.3.8 High-impact shock test. -

4.3.8.1 One complete compressor unit of each type, design, and size of classes CC and S shall be selected by the Government inspector for type A shock tests in accordance with Specification MIL-S-901.

4.3.8.1.1 Only one compressor of each type, design, and size shall be subjected to the shock tests, and the tests need not be repeated on later contracts or orders unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Changes in materials of construction may require retesting. Changes in the driving motor only shall not be considered a reason for retesting a compressor; however changes in the motor will require shock testing of the motor.

4.3.8.1.2 Consideration will be given to waiving the shock test requirement on a compressor unit providing the design of that compressor is practically identical to that of another unit which has successfully passed the shock test. The decision as to whether a shock test will be waived shall rest with the bureau or agency concerned.

4.3.8.1.3 If a manufacturer is unable to conduct the shock tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory suitably equipped to conduct the tests. When the shock tests are to be conducted at a Government laboratory a complete set of master and outline drawings shall accompany the equipment.

4.3.8.2 The compressor shall be idle and running on alternate blows of the shock test, but the compressor shall not be compressing air while running.

4.3.8.3 Inspection after shock test. -

4.3.8.3.1 Motor. - Inspection of the motor shall be in accordance with Specification MIL-M-17060 or MIL-M-17413.

4.3.8.3.2 Compressor. - Inspection of the compressor shall be conducted as follows:

(a) Examine the unit carefully and note the extent of any external damage.
(b) Jack the compressor over several times by hand to insure that all working parts are free and ready for starting.
(c) Run the compressor unloaded for a short time and note any mechanical peculiarities.
(d) Gradually build up the load and carefully check any undue noises or excessive vibration or other nonconformances.
(e) Secure the compressor and disassemble for complete inspection. Observe and record all effects of the shock test on the various compressor components.

4.3.8.4 Failure to pass test. -

4.3.8.4.1 Motor. - The motor will be considered to have failed the shock test unless it conforms to Specification MIL-M-17060 or MIL-M-17413.

4.3.8.4.2 Compressor. - The compressor will be considered to have failed the shock test in the event of the following:

(a) Breakage of any essential parts, including mounting, bolts.
(b) Detachment of any part or object from the unit.
(c) Compressor requires attention or adjustment or replacement of parts to enable it to perform its principal function during and after each blow of the test.
(d) Unit fails to function satisfactorily as specifically defined in this specification.

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4.3.3.5 Disposition of shock-tested equipment. - Equipment which has successfully passed the shock tests will be considered acceptable for such service as designated by the bureau or agency concerned. The contractor shall thoroughly recondition such units, correcting any minor imperfections or deformations, and shall place the equipment in satisfactory working order. The motors shall be reconditioned and tested in accordance with the requirements of the motor specifications. After reconditioning, the equipment shall be tested as specified in 4.3.3.2.

4.3.3.5.1 Equipment which has been subjected to the high-impact shock tests and has failed will not be accepted, either in whole or in part until the causes of such failure have been corrected by the manufacturer, and the complete equipment has been successfully retested. However, failure of the compressor only shall not cause rejection of an otherwise acceptable motor, nor shall failure of a motor only cause rejection of an otherwise acceptable compressor.

4.3.9 Test reports. -

4.3.9.1 Test reports shall be prepared for all tests conducted by the contractor, including the shock and inclination tests. Recorded data of the compressor performance tests may be logged on contractor's forms. The following identification data for each compressor unit shall be furnished and included in the test report:

(a) Serial numbers of the compressor and driving unit.
(b) Ship for which intended.
(c) Contract or order number.
(d) Actual net weight of the unit in the "dry" and "wet" condition.

4.3.9.2 Distribution of compressor performance test reports shall be as follows:

(a) Two to the bureau or agency concerned.
(b) Two to each supervisor of shipbuilding for each ship's worth of compressors furnished (one for Supervisor's files; balance for shipbuilder) for privately built ships.
(c) Two to each Naval Shipyard for each ship's worth of compressors for Navy built ships.
(d) One for primary inspection officer.
(e) One for branch inspection office (if any).

4.3.9.3 Distribution of shock test reports and inclination test reports shall be as follows:

(a) Six to the bureau or agency concerned, for approval.
(b) One for the primary inspection office.
(c) One for the branch inspection office (if any).

5. PREPARATION FOR DELIVERY

5.1 General. - Equipment and repair parts shall be prepared for delivery in accordance with the following requirements as specified (see 3.1):

Levels A and B - Domestic shipment and storage or overseas shipment.
Level C - Domestic shipment - immediate use.

5.2 Preservation and packaging. -

5.2.1 Level A. - Cleaning, preservation and packaging methods and materials shall be in accordance with Specification MIL-P-116.

5.2.1.1 Compressor. - Internal surfaces of the compressor shall be flushed with preservative type P-10. Exterior, unpainted surfaces subject to corrosion shall be coated with preservative type P-2 and wrapped or covered with greaseproof barrier material conforming to grade A of Specification MIL-B-121 secured in place with tape as necessary. Openings to the interior of the compressor shall be sealed with tape conforming to grade B of Specification JAN-P-127 or barrier material conforming to Specification MIL-B-121 secured in place with tape

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5.2.1.2 Coolers. - Intercoolers and aftercoolers shall be dried by passing dry air through the system. All openings in the coolers shall be sealed.

5.2.1.3 Motor. - The motor shall be preserved and packaged in accordance with method II of Specification MIL-P-116. If the mounting does not permit obtaining a water-vaporproof seal as required in method II, the alternate method of sealing the motor as specified in Specification MIL-P-16298 may be used.

5.2.1.4 Reduction gears. - Reduction gears shall be flushed with preservative type P-2.

5.2.1.5 Instruments panel board. - Instruments not removed from the basic unit shall have all openings sealed with pressure sensitive tape conforming to type I, grade B of Specification JAN-P-127. Unless otherwise protected against breakage, dial glasses shall be masked with masking tape, covered with barrier material, in accordance with type I, grade A, class 1 of Specification MIL-B-121, then protected with pieces of plywood or fiberboard secured in place with pressure sensitive tape.

5.2.1.6 Controllers. - Detached controllers and detached electrical components shall be preserved and packaged in accordance with Specifications MIL-C-2174, MIL-C-2212 and MIL-P-17652, as applicable.

5.2.1.7 Repair parts and tools. -

5.2.1.7.1 Electrical repair parts and tools shall be preserved and unit packaged in accordance with Specifications MIL-P-17652 and MIL-P-16298, and in accordance with Specification MIL-P-116 for items not specifically covered therein.

5.2.1.7.2 Mechanical repair parts shall be preserved and unit packaged in accordance with Publication JPI-6 and in accordance with Specification MIL-P-116 for items not specifically covered therein.

5.2.1.7.3 When specified (see 6.1), onboard repair parts shall be packaged as a set in onboard repair parts stowage boxes conforming to type M or W of Specification MIL-B-233 as specified. The hardware of type W shall be of nonmagnetic materials exclusively. The gross weight of the boxes shall not exceed 200 pounds.

5.2.2 Level C. - The compressor shall be cleaned, preserved and packaged in accordance with commercial practice.

5.3 Packing. -

5.3.1 Level A and level B. -

5.3.1.1 Components removed from the basic unit to facilitate shipment, which require segregation or protection from mechanical damage, shall be packaged in nailed wood boxes conforming to Specification NN-B-621 or in nailed wood cleated boxes conforming to Specification PPP-B-601.

5.3.1.2 The complete unit, including detached components, shall be anchored to the base of a nailed wood sheathed crate conforming to Specification MIL-C-3769 or JAN-P-104, as applicable. Blocking, bracing and cushioning shall be in accordance with Specification JAN-P-100. The crates shall be ventilated in the manner specified in Specification JAN-P-104.

5.3.1.3 Repair parts and tools. - Repair parts and tools shall be packed by level A or level B as specified (see 6.1) in accordance with Specification MIL-P-17652. An index list conforming to Specification MIL-F-17292 shall be secured within the onboard repair parts container.

5.3.2 Level C. -

5.3.2.1 The equipment and related accessories (except repair parts and tools) packaged as specified in 5.2.1 shall be packed in a manner which will insure acceptance and safe delivery at destination. The containers shall comply with the Consolidated Freight Classification Rules or Motor Freight Classification Rules, whichever may be applicable.

5.4 Publications. - Technical manuals shall be packaged and packed in accordance with Specification MIL-T-15071.

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5.5 Drawings. - Drawings shall be prepared for shipment in accordance with Specification MIL-D-960.

5.6 Marking. - In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size required (see 1.2).
(c) Floor space, height and weight limitations, if any. Limits of height, length and width for disassembly if any (see 3.3.1.1 (f)).
(d) Method of cooling of class R and S compressors (see 3.3.7.16.1).
(e) When compressors may have air-cooled cylinder heads (see 3.3.7.16.2.2).
(f) Voltage and current characteristics required (see 3.5.1 (b)).
(g) Repair parts required (see 3.6).
(h) Quantity of compressors installed aboard a vessel (see 3.6.2.2).
(i) Tool and wrench requirements (see 3.6.3).
(j) Selection of applicable levels of preservation, packaging and packing required (see 5.2 and 5.3).
(k) Whether onboard repair parts boxes are required; if so, specify whether type M or W (see 5.2.1.7.3).

6.2 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-19113, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 Heat rejection guide. - For estimating heat rejection from an air cooled compressor, in determining whether an air cooled compressor would be satisfactory in a specific location, the following may be assumed as a guide. The percentages given are approximate only and are not intended as a design or test requirement.

15 percent of compressor brake horsepower from the cylinders.
40 percent of compressor brake horsepower from the coolers. If the compressors are two stage machines the heat rejection from the coolers may be divided as follows:
  15 percent of compressor brake horsepower from the intercoolers.
  25 percent of compressor brake horsepower from the aftercoolers.
1 brake horsepower = 42.41 BTU per minute.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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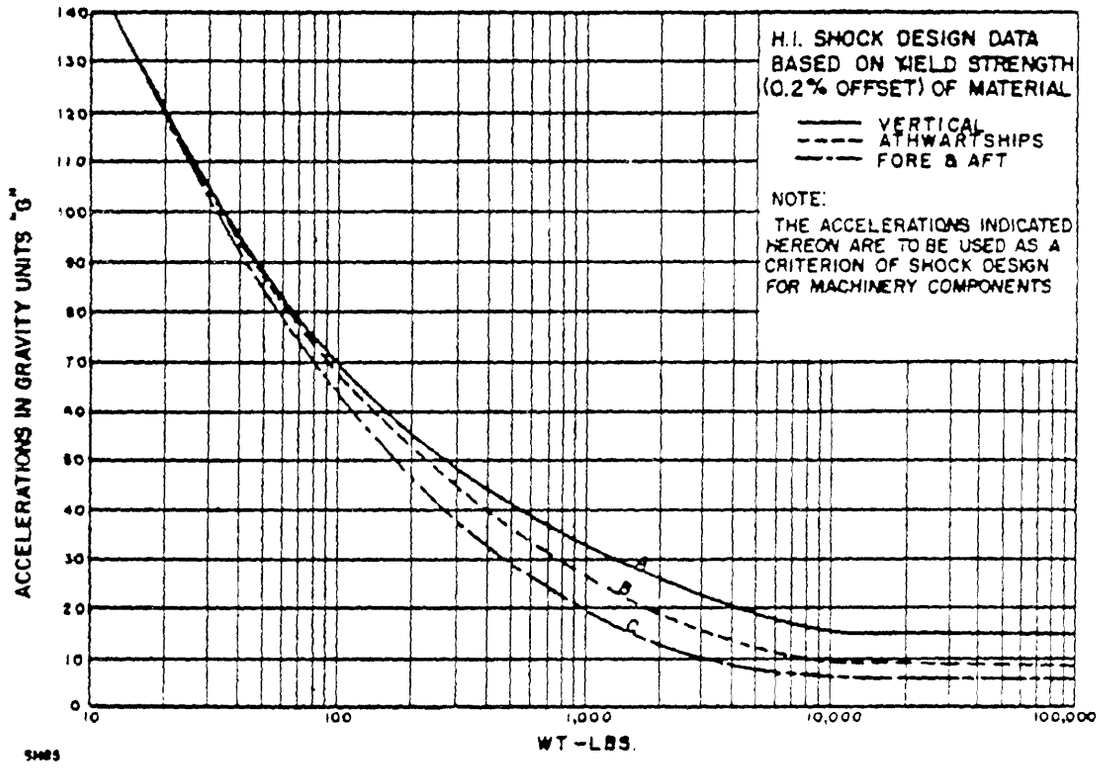


Figure 1. – H.I. Shock design data.

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INDEX

	Referenced paragraph	Page
Aftercoolers	3.3.7.16.3.5	9
Air cooled	3.3.7.16.2	8
Air cylinders and heads	3.3.7.8	7
Air filters	3.3.7.11	7
Air intake	3.3.3	6
Air passages	3.3.7.3	7
Air temperature, ambient	3.3.3	6
Air temperature, discharge	3.3.3	6
Applicable documents	2	2
Assembly	3.3.1	5
Bearings	3.3.7.2.2	9
Bedplates	3.3.7.9	7
Capacity test	4.3.6.1.1.	19
Check valves	3.3.1.1	5
Classes	1.2	1
Computed data	4.3.6.1.3	20
Conditions, design	3.3.3	6
Connecting rods	3.3.7.19	9
Controllers, electric	3.5.2	14
Coolers	3.3.7.16.3.5	9
Cooling systems	3.3.7.16	8
Cooling system valves	3.5.2.2	15
Cooling water	3.3.3	6
Crankshaft	3.3.7.20	9
Cylinders	3.3.7.8	7
Data, ordering	6.1	24
Data, recorded	4.3.6.1.2	19
Delivery, preparation for	5	22
Description	3.3	5
Design conditions	3.3.3	6
Design, general	3.3	5
Disposition of shock-tested equipment	4.3.8.5	22
Distribution	3.6.2	18
Documents	2	2
Drain valves	3.3.7.4	7
Drawings	3.7	17
Drive	3.3.4	6
Electrical equipment	3.5	14
Electrical test	4.3.5	19
Filters air	3.3.7.11	7
Flywheels	3.3.7.21	9
Gageboard	3.3.7.4	7
Gages	3.3.7.18	9
Heads	3.3.7.8	7
Heat rejection guide	6.3	24
Housing	3.3.7.10	7
Humidity	3.3.3	6
Hydrostatic test	4.3.4	19
Inclination test	4.3.7	20
Identification plates	3.3.10	10
Jacking device	3.3.7.15	8
Lot acceptance tests	4.3.2	18
Lubrication	3.3.7.12	7
Lubricating oil coolers	3.3.7.12.3	7

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INDEX

	Referenced paragraph	Page
Manuals, technical	3.8	18
Marking	5.6	24
Master drawings	3.7.3	17
Master switches	3.5.2	14
Materials	3.4	10
Moisture separators	3.3.7.23	10
Motors	3.5.1	14
Noise	3.3.7.6	7
Notes	6	24
Onboard repair parts	3.6.2	15
Ordering data	6.1	24
Outline drawings	3.7.2	17
Overseas shipment	5.1, 5.2 and 5.3	22, 23
Packing	5.3	23
Painting	3.3.9	10
Performance tests	4.3.6	19
Piping and valves	3.3.7.13	8
Piston speed	3.3.7.2	7
Plates, identification	3.3.10	10
Preparation for delivery	5	22
Preservation	5.2	22
Qualification requirement	3.1, 6.2	5, 24
Qualification test	4.3.3	18
Quality assurance provisions	4	18
Radio noise	3.5.1.2	14
Rating	3.3.5	6
Recorded data	4.3.6.1.2	19
Relative humidity	3.3.3	6
Repair parts	3.6	15
Repair parts, stock	3.6.4	17
Reports, test	4.3.9	22
Rods, connecting	3.3.7.19	19
Sampling	4.2	18
Scope	1.1	1
Sectional assembly drawings	3.7.3	17
Selector switch	3.5.2.1	14
Separators, moisture	3.3.7.23	10
Shockproofness	3.3.6	6
Shock tests	4.3.8	21
Speed, piston	3.3.7.4	7
Standard product	3.3.2	6
Subassemblies, drawings	3.7.4	18
Switches, master	3.5.2	14
Switches, pressure	3.5.2.1.4	15
Technical manuals	3.8	18
Temperature, air	3.3.3, 3.3.7, 16.1	6, 8
Temperature, water	3.3.3	6
Tests, hydrostatic	4.3.5	19
"Test" push buttons	3.5.2.1.3	15
Test reqport	4.3.9	22
Test, qualification	4.3.3	18
Test, shock	4.3.8	21
Thermometers	3.3.7.24	10
Threaded parts	3.3.7.14	8
Tools	3.6.3	16

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Check the source to verify that this is the current version before use.

INDEX

	Referenced paragraph	Page
Unloading	3.3.7.17, 3.5.2.1.2, 3.5.2.2	9, 16
Valves, check	3.3.1.1	5
Valves, cooling system	3.5.2.2	15
Water, cooled	3.3.7.16.3	8
Water, cooling	3.3.3	6
Welding	3.3.8	10
Wrenches	3.6.3	16
Zincs	3.3.7.16.3.7	9

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